Automated Fluorescence Image Analysis of RNA Condensates

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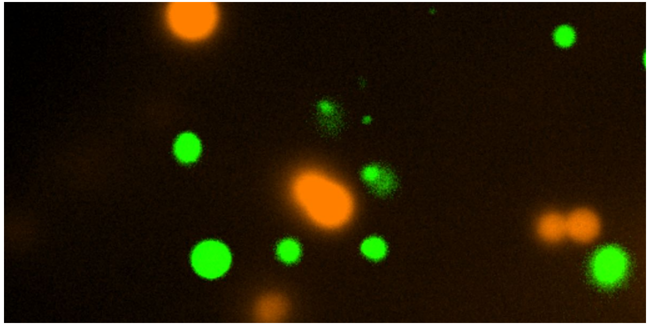

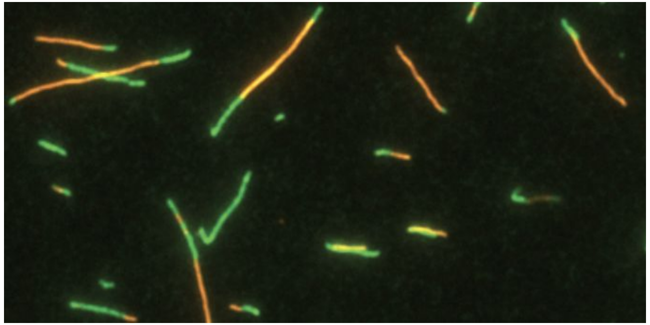

RNA Condensates

- **Condensates**: Dense, membrane-less droplets forming when molecules exceed a concentration

- Synthetic RNA condensates using programmable **RNA nanostars**
    - 100-200 nt single strands folding into multiarm structures and interact via kissing loops

- **Design Parameters**: arm length, number of arms, kissing loop sequence control

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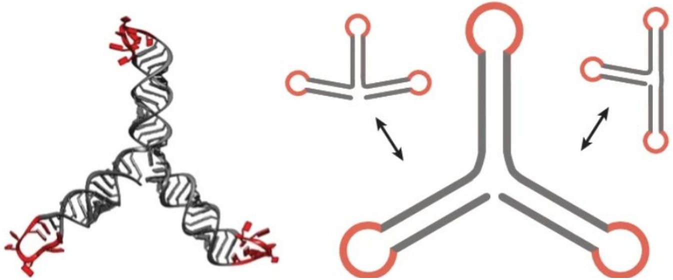
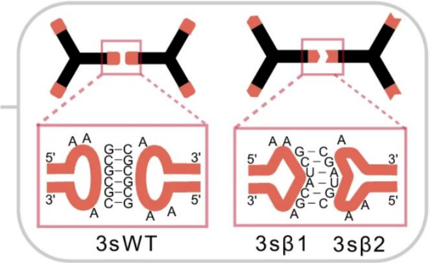


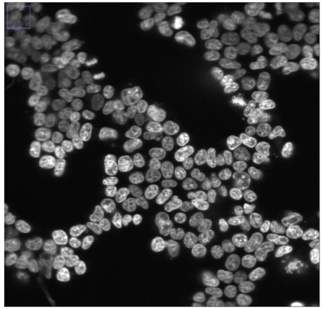


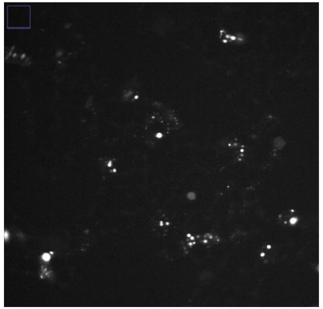
The JABr Construct: Dataset Explanation

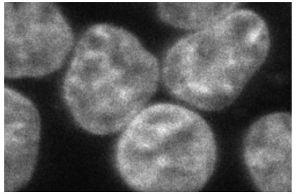
- **JABr** = 15-nt stem, 3-arm, kissing-loop variant A (UCGCGA), Broccoli fluorescent aptamer

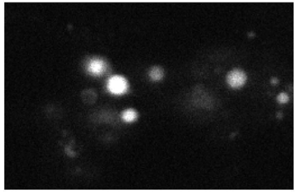
- 15-nt arm arm length = **nuclear pore permeability threshold** - forms in both nucleus and cytoplasm

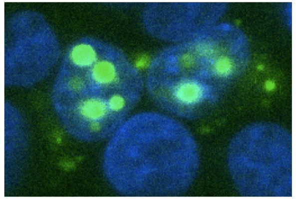
- **Partition Coefficient (PC)** = Phase Separation Strength
    - Condensate intensity inside nucleus / intensity outside











The Problem: Manual Analysis Doesn’t Scale

- **Existing analysis** was Manual, Slow, and Hard to reproduce
  - Prior: Manually check and segment nuclei and condensates using ImageJ/Fiji
- **Goal:** Automated Python pipeline taking in raw-stacks as inputs and outputs segmentation masks, measurements, and PCs

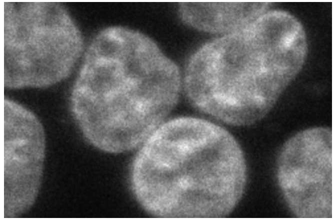
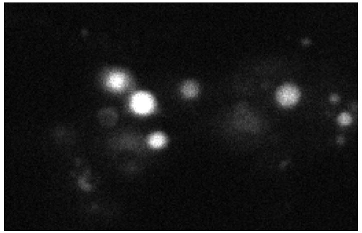
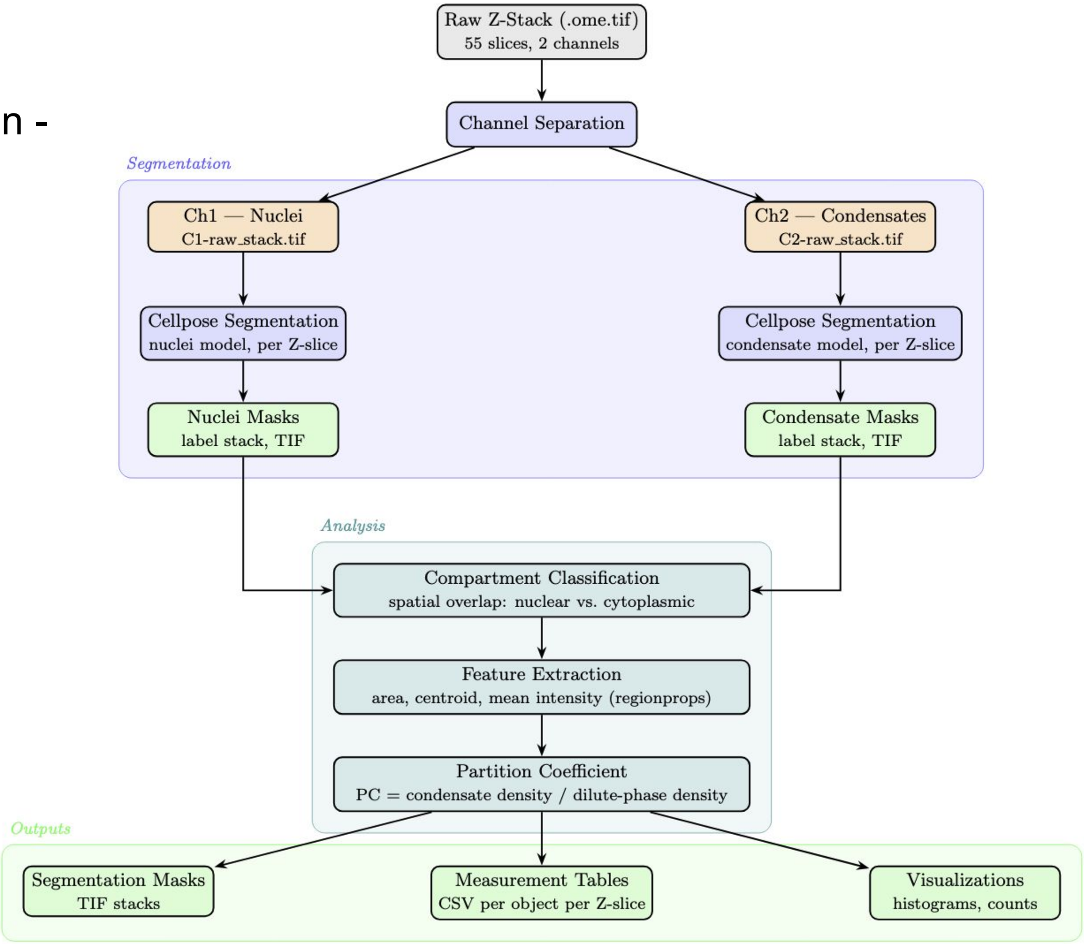



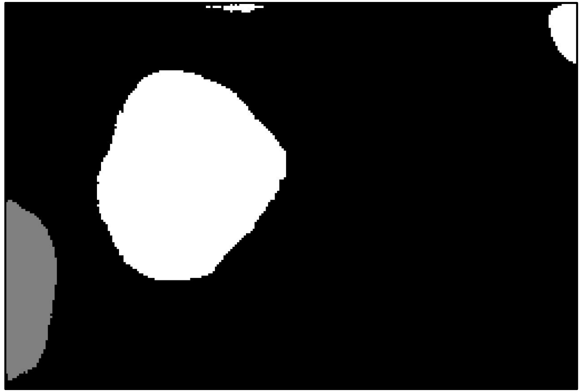
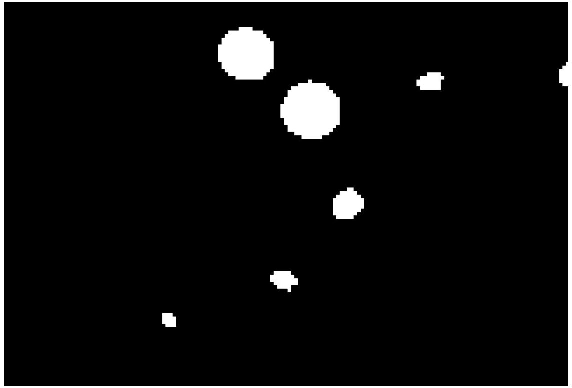
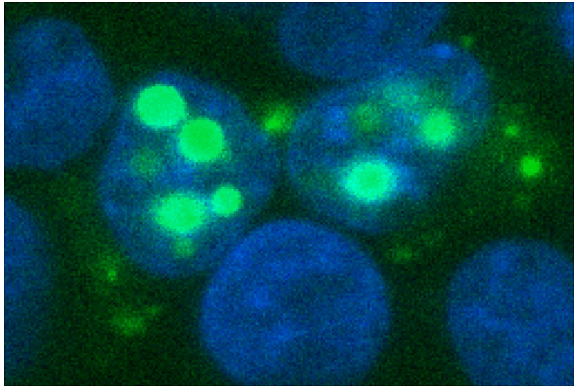
Image Analysis Pipeline

Cellpose Segmentation - scikit-image feature extraction - Python



Quantifying Nuclear Enrichment

Figure 8

Nuclei Mask (Ch1) (z = 22)	Condensate Mask (Ch2) (z = 22)	Raw Fluorescence (z = 22)
		
Which pixels are nucleus	**Which pixels are condensate**	**Intensity values measured**

- **PC** = condensate density / dilute phase density
- **Condensate density** — mean intensity of pixels inside nucleus ∩ condensate mask
- **Dilute-phase density** — mean intensity of pixels inside nucleus, outside condensate mask
- **PC > 1** → condensates are enriched relative to surrounding nucleoplasm